



Public Perceptions of Police Use of Force Encounters: Demographic, Attitudinal, and Situational Predictors

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Abstract

The current work builds on past research by examining the extent to which demographic, attitudinal, and situational factors predict public perceptions of police use of force. Specifically, we move beyond general perceptions of force into specific perceptions of threat and responsibility related to both the officer and civilian. In a between-subjects experimental design, participants ($N=724$) listened to a traffic stop vignette where a civilian engages in varying levels of noncompliance (less noncompliance v. more noncompliance) and the officer uses force. Participants completed measures of threat, responsibility, and reasonableness, as well as demographics and attitudes towards police legitimacy and authoritarianism. Participant age, education level, and political orientation significantly impacted judgments. Yet, the impact of demographics was diminished when police legitimacy and authoritarian attitudes were accounted for. The situational influence of civilian action also predicted perceptions, with attitudinal influences remaining significant even when controlling for civilian noncompliance. Individuals' education level and police legitimacy attitudes predicted their perceptions of initial threat to both the officer and civilian prior to any officer force or civilian noncompliance. Implications and future directions are discussed.

Keywords Authoritarianism · Police · Police legitimacy · Public perceptions · Threat · Use of force

In 2015, the United States (U.S.) Federal Bureau of Investigation (FBI) established the “National Use of Force Data Collection” initiative to create a national-level dataset on police use of force incidents (FBI, n.d.). In 2024, data from the initiative indicated 55% of reported police-civilian incidents (by approximately 60% of U.S. policing agencies) involved serious bodily harm to a civilian and 33% involved a civilian death (Federal Bureau of Investigation 2024). Recent and continued media coverage and public outcry surrounding these incidents (e.g., Bogel-Burroughs and Fazio 2021), opinion polls of perceptions of police (e.g., Martin and Higginbotham 2024; Morin et al. 2017), and

research into what factors may predict the public's perception of police use of force (e.g., McClure et al. 2023; Vardaveen & Wiener, 2022) all suggest an overarching desire to understand police-civilian interactions better.

Gaining an accurate understanding of the public's perception of police-civilian encounters is important for several reasons. First, individuals' perceptions of police-civilian encounters have been found to predict the extent to which individuals will cooperate with police when they themselves interact with the police (Jonathan-Zamir et al., 2015; Tankebe 2009; Tyler and Huo 2002). This is notable given that over 50 million people encountered police in the U.S. in 2020 alone (Tapp and Davis 2022). Second, perceptions of police-citizen interactions are also related to individuals' attitudes about the legitimacy of the police (Tyler 2001, 2003). Perceptions of police legitimacy predict individuals' willingness to obey the law more generally, with more positive attitudes associated with a greater obligation to law-abiding behavior (Tankebe 2013; Tyler and Fagan 2008). Third, understanding public perceptions of police-civilian encounters (particularly those involving the use of force) is vital from a legal decision-making perspective. In

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the U.S. legal system, citizens serve as jurors in criminal and civil cases of alleged excessive use of force by police officers. Juror decision-making research has found that jurors' case-related perceptions (e.g., threat to the officer, civilian injury) and attitudes influence their verdicts and related legal judgments (e.g., Chapman and Bornstein 1996; Estrada-Reynolds et al. 2015; Sivasubramaniam et al. 2020). Jurors in alleged excessive use of force cases must determine the extent to which the officer's use of force aligns with the level of force any reasonable police officer on scene at the time would have used (i.e., the reasonable officer standard; *Graham v. Connor* 1989). Limited research examining the reasonable officer perspective (Vardsvén & Wiener, 2022) indicates the importance of mock jurors' perceptions of threat and resistance on their reasonable officer standard-related judgments. Therefore, understanding the public's perceptions of police-civilian encounters can provide insight into how such cases may be decided in court. While prosecution of police officers for use of force is infrequent (Stinson, 2019), continued public attention and outcry from highly publicized use of force incidents (e.g., Pitts 2022) may increase cases in the future.

In attempts to further understand police-civilian encounters, researchers have begun to examine how perceiver demographics (e.g., Navarro and Hansen 2023) and attitudes (e.g., Ewanation and Maeder 2023), as well as situational encounter-specific factors like the level of civilian aggression (e.g., McClure et al. 2023) or a civilian's behavior (e.g., Flick and Schweitzer 2025), can predict public perceptions of police use of force. The existing literature (discussed in detail below) regarding the role of demographics (e.g., political affiliation, age, and sex) and attitudes (e.g., police legitimacy) has primarily focused on how the perceiver's demographics or attitudes predict perceptions of whether the force used by officers was reasonable (compared to excessive). Understanding how individual differences, such as demographics or attitudes, impact use of force judgments may benefit attorneys in the jury selection process for alleged excessive use of force cases. Research suggests that jurors' case-related attitudes (e.g., Chapman and Bornstein 1996; Estrada-Reynolds et al. 2015; Sivasubramaniam et al. 2020) and demographics (albeit to a lesser extent; Lieberman and Sales 2007) impact their legal judgments. Beyond the courtroom, a knowledge of how demographic characteristics impact public perceptions of police and police use of force has the potential to improve police training and improve communication between officers and civilians in various communities. For instance, knowing how different demographics perceive the police and police-civilian interactions may improve the cultural sensitivity training some police agencies provide, for which no current nationally centralized training exists (Moon et al. 2018).

While reasonable force judgments are important and directly related to how jurors may decide relevant legal cases, a more nuanced examination into how these factors predict perceptions of other facets of police use of force encounters, such as perceptions of threat, is essential. This is particularly true given that scholars have recently called for a greater emphasis on understanding threat within police-civilian encounters (Bennell et al. 2021). Given that research has shown a civilian's behavior can impact how threatening the situation is perceived to be (e.g., McClure et al. 2023), we sought to examine how a perceiver's demographics and attitudes, along with the situational influence of a civilian's behavior, affect the public's perceptions of threat in a police-civilian use of force encounter. In the current work, we experimentally manipulated the civilian's action in a police use of force encounter. We also measured participants' perceptions of police legitimacy, authoritarianism, and demographic information to examine how each predicted their perceptions of threat in a police-civilian interaction.

Demographic Predictors of Perceptions of Police Use of Force

When examining public perceptions across a variety of criminal justice issues, an individual's demographics can play an important role in predicting their judgments. A handful of demographics previously examined within perceptions of policing include, but are not limited to, race, sex, age, political orientation, education, and socioeconomic status. When examining perceptions of police use of force encounters specifically, the role of these demographics is far from conclusive. Some previous research finds demographics like sex and age play an important role in predicting perceptions of use of force interactions (e.g., Simon et al. 2021; Trahan and Russell 2017), and other studies find no impact, or opposite effects (e.g., Mourtgos and Adams 2020; Weitzer and Tuch 2006). In this section, we briefly review how a perceiver's race, sex, age, political orientation, education level, and income affect their perceptions of police and use of force incidents.

Race

Race and ethnicity are commonly studied factors in police use of force (e.g., Arthur and Case 1994; Brown and Benedict 2002; Dierenfeldt et al. 2024; Mourtgos and Adams 2020; Roche and Roux 2017). Previous research consistently indicates that racial minorities have more negative views of police use of force compared to White individuals (e.g., Boudreau et al. 2019; Jefferis et al. 2011; Johnson

and Kuhns 2009; Peffley and Hurwitz 2010; Strickler and Lawson 2020; Wu 2014). For instance, Boudreau and colleagues (2019) examined public perceptions of police use of deadly force and how these perceptions impact how individuals assign blame for these incidents across various related actors. Results indicated that these judgments varied by race, such that Black respondents, compared to non-Black respondents, attributed significantly more blame to police during these deadly force incidents when respondents were given the choice of assigning blame to the police, the victim, or local or state/federal officials. Increased disapproval of police use of force is not limited to comparing White and Black individuals, with research showing Latinos and Asian Americans also express more unfavorable views of police use of force compared to White Americans (e.g., Berk-Seligson 1990; Carter 1983; Sun & Wu, 2015). This trend is arguably unsurprising given the history of police mistreatment of racial minorities in the United States (e.g., Hadden 2021; Hurwitz and Peffley 2005; Johnson and Kuhns 2009).

Sex

Contrasting race, previous research on sex differences in police use of force attitudes finds inconsistent results. Some research suggests men generally express higher approval of police use of force (across reasonable and excessive force scenarios) compared to women (Arthur and Case 1994; Barkan and Cohn 1998; Johnson and Kuhns 2009; Mourtgos and Adams 2020; Ratcliffe et al. 2015; Simon et al. 2021; Thompson and Lee 2004). However, other research has found women are more in favor of police use of excessive force than men, and suggest no sex differences when the force used was reasonable (Cao et al. 1996; Gerber & Jackson, 2017; Lai and Zhao 2010; Navarro and Hansen 2023; Reisig and Correia 1997; Weitzer and Tuch 2002). Other studies reported no relationship between sex and satisfaction with police more generally (Haberman et al. 2016; Weitzer and Tuch 2006).

Age

Age is another demographic factor that can influence perceptions of police use of force, with research indicating somewhat mixed results. When considering general attitudes towards police, older individuals tend to report greater satisfaction with and less suspicion of police than younger adults (e.g., Brown and Coulter 1983; Cao et al. 1996; Chermak et al. 2001). When looking at police use of force judgments specifically, research suggests older individuals tend to be more approving of police use of force than younger individuals (e.g., Trahan and Russell 2017), while other research suggests no age differences (e.g., Carter and Corra

2016; Mourtgos and Adams 2020; Simon et al. 2021). One potential explanation for these contradictory results is that the impact of age may be dependent on just how excessive the force used by police was. Older adults tend to be less in favor of police use of excessive force and more in favor of reasonable force compared to younger adults (Gerber & Jackson, 2016; Navarro and Hansen 2023).

Political Orientation

When considering the impact of an individuals' political orientation, previous research indicates those with more liberal (compared to conservative) political views tend to hold more unfavorable opinions of police (e.g., Cao and Stack 2005; Cao and Zhao 2005; Tyler et al. 1997) and their use of force (e.g., Braga et al. 2014; Mourtgos and Adams 2020; Trahan and Russell 2017). Research also shows liberal-leaning individuals tend to favor punishing officers who have committed a false-positive error (i.e., shooting an unarmed person) to a greater extent than more conservative individuals (Patil 2018). This relationship between political orientation and views of police is not unique to the U.S.; research from the United Kingdom also finds an association between political conservatism and increased positive perceptions of police use of force (Bradford et al. 2017). Notably, recent work by Navarro and Hansen (2023) suggests a potentially more nuanced view of liberal individuals' perceptions. Liberal individuals' views of police use of force appear to be more dependent on the threat level present in the use of force scenario, when compared to conservative individuals. Conservatives tended to approve of police use of force when force was used when there was a violent threat to others or the officer, and conservatives maintained strong approval for force across situations that lacked a clear threat of violence, aligning with previous research (e.g., Patil 2018; Tyler et al. 1997). In contrast, liberals' judgments varied as a function of the threatening situation, such that liberals' support for the use of force decreased as threat levels became less clear.

Education

The empirical evidence on the impact of education on public use of force judgments is mixed. Research shows that individuals with higher levels of education tend to express greater approval for police use of force (e.g., Arthur and Case 1994; Halim and Stiles 2001; Silver and Pickett 2015; Simon et al. 2021; Trahan and Russell 2017), whereas individuals with less formal education tend to hold more negative perceptions of the police, particularly if they have personally been exposed to police violence (Jackson et al. 2021). However, some studies suggest a more complex relationship, indicating education's impact might depend

on situation-specific factors. Navarro and Hansen (2023) found that the impact of education on use of force judgments may depend on the officer's specific actions and level of force, with higher levels of education being associated with greater approval of police use of excessive, but not reasonable, force. Lastly, some studies suggest either no significant relationship between education and perceptions of police use of force (Johnson and Kuhns 2009) or an inverse relationship (as education increases, approval decreases; Braga et al. 2014; Mourtgos and Adams 2020). For instance, people with baccalaureate degrees or higher (compared to those with lower educational degrees) have been found to have more negative general assessments of police officers' actions (Braga et al. 2014).

Income

Income is another factor considered within the context of public perceptions of police and use of force. Overall, empirical evidence supports the notion that individuals with higher income tend to be more approving of police and police use of force (e.g., Brown & Cutler, 1983; Cao et al. 1996; Carter and Corra 2016; Mourtgos and Adams 2020; Parker et al. 1995; Simon et al. 2021). For instance, Braga and colleagues (2014) assessed individuals' income levels and general attitudes toward police. Respondents with incomes over \$40,000 had significantly more positive attitudes toward police than those with less than \$40,000. Jackson and colleagues (2021) reported similar correlations across police legitimacy/trust, effectiveness, and satisfaction. This trend is supported by a 2016 national survey that found 79% of white Americans with incomes over \$60,000 held more favorable attitudes toward police compared to 56% of white Americans with incomes of less than \$30,000 (Ekens 2016). This relationship is seen with perceptions of police use of force as well, with numerous studies indicating that individuals with higher income are more approving of officers using force (e.g., Mourtgos and Adams 2020; Simon et al. 2021).

Summary of Demographic Predictors

Previous research regarding individual demographic factors of race, political orientation, and income on individuals' perceptions of police and police use of force has shown largely consistent results. Findings suggest that individuals who are racial minorities, hold liberal political views, or have lower incomes tend to have more negative perceptions of policing and use of force. In contrast, the influence of an individual's sex, education level, and age on these judgments has been generally inconsistent. There are several potential explanations for these inconsistent findings. First, methodological

variations across studies can lead to different outcomes (e.g., Green and Peneff 2022). Second, cultural, situational, and contextual elements often influence individual perceptions and responses. These, and other factors, can make it difficult to draw general conclusions regarding the effect of perceiver demographics on perceptions of police use of force incidents. The current study aims to better understand how an individual's sex, age, political orientation, education, and income level impact their perceptions of threat in a police use of force scenario.

Attitudinal Predictors of Perceptions of Police Use of Force

In addition to individuals' demographic characteristics, some attitudinal measures have also been found to affect public perceptions of police use of force incidents. Contrary to the handful of demographics discussed above, certain attitudes predict individuals' judgments more consistently: police legitimacy beliefs and authoritarian views.

Attitudes Toward Police Legitimacy

A key individual difference found to be important when assessing people's perceptions of police-civilian interactions is the extent to which they believe the police are a legitimate authority in society. Police legitimacy comprises multiple dimensions, including procedural fairness, distributive fairness, lawfulness, and effectiveness (Tankebe 2013). Legitimacy directly influences citizens' willingness to cooperate with police and indirectly affects cooperation through perceived obligations to obey (Tankebe 2013; Tyler and Fagan 2008). Procedural justice, characterized by citizen participation, perceived neutrality of the authority in their decision-making, respect by the authority, and trustworthiness of the authority, is a key factor in shaping public perceptions of police legitimacy (Tyler 2004; Mazerolle et al. 2013). Research indicates that experiencing procedural justice during personal encounters with police enhances legitimacy, regardless of outcome favorability (Tyler and Fagan 2008).

Based on Tyler's (1990, 2006) work on why people obey the law, Reynolds and colleagues (2018) developed a scale to assess people's attitudes toward police legitimacy. The original scale, or its validated short form (Reynolds 2023), has been used in many studies exploring perceptions of police-civilian interactions, and it consistently predicts individuals' decisions (e.g., Ewanation and Maeder 2023; Smith and Nuñez 2024). People with greater beliefs that the police are a legitimate authority tend to perceive police use of force incidents as more justified (e.g., Reynolds et al. 2018),

officer's use of force actions as more acceptable (Celestin and Kruschke 2019), be less likely to say an officer should be prosecuted for murder (Burd et al. 2025), be less likely to convict an officer who used force (e.g., Ewanation et al. 2022), and be less likely to suggest punishment for an officer who used force (Celestin and Kruschke 2019; Flick and Schweitzer 2025). Further, individuals with particularly low legitimacy beliefs rarely view lethal police UOF as acceptable, even when a civilian uses lethal levels of force against an officer (Celestin and Kruschke 2019).

Authoritarianism

Authoritarian individuals tend to be considered submissive to authority, hostile towards those who do not follow social norms, and guided by traditional values and social conventions (Altemeyer 2004; Passini 2017). Recent research challenges the traditional conceptualization of right-wing authoritarianism by Altemeyer (2004), proposing a three-factor model comprising authoritarian submission, authoritarian aggression, and conventionalism instead. The Aggression-Submission-Conventionalism (ASC) scale, developed by Dunwoody and Funke (2016), is a three-factor measure of authoritarianism that improves upon Altemeyer's Right-Wing Authoritarianism scale (Altemeyer 2004) and has been shown to better capture authoritarianism than one-dimensional measures (Funke 2005; Dunwoody and Funke 2016).

Importantly, authoritarianism has been shown to be related to police legitimacy and perceptions of use of force. People with higher authoritarian views are more likely to view the police as legitimate (Reynolds et al. 2018), suggesting the measures are related but distinct constructs. Individuals high in authoritarianism have also been found to view forceful actions by police more positively (e.g., Watson and Stevenson 2022) and to view police excessive use of force as justified (e.g., Milani 2020), even when controlling for police legitimacy attitudes (Gerber and Jackson 2017). Those with higher, compared to lower, authoritarian beliefs have also been shown to believe police officers who use force should receive less professional and legal punishment (Flick and Schweitzer 2025). They also have more positive perceptions of the officer and more negative perceptions of the civilian involved in the use of force encounter (Flick 2023). Lastly, increased authoritarian beliefs are associated with increased perceptions of societal threat (e.g., crime rates, economic threat; Doty et al., 1991; Roccato et al. 2014). Given previous research on perceptions of police use of force and the importance threat perceptions play in authoritarian worldviews, it seems possible that authoritarian beliefs may be particularly predictive of individuals' perceptions of threat in a police use of force encounter.

Situational Predictors of Perceptions of Police Use of Force

Previous research suggests that various situational factors predict whether, and the extent to which, an officer uses force in a police-civilian encounter. Some of these situational factors include but are not limited to suspect resistance, suspect intoxication, presence of a weapon, location of the encounter, number of citizen bystanders, crime seriousness, and number of other officers present at the scene (e.g., Engel et al. 2000; Kavanagh, 1997; Moyer 1981; Scharf and Binder 1983; Wolf et al. 2009). However, how the public perceives these different situational predictors of an officer's use of force is less studied. For instance, Simpson (2017) found that the presence of a police uniform (compared to plain clothes) increased participants' perceptions of an officer as aggressive, and foot patrol and bicycle officers were rated more positively (e.g., approachable, respectful) than vehicle patrol officers.

One notable situational factor explored across both police use of force decisions and public perceptions of force is the level of suspect resistance, which appears to be the most reliable predictor of whether an officer will use force, or the level of force used (e.g., Boivin 2016; Engel et al. 2000; Garner and Maxwell 1996; Kavanagh 1997; McLean et al. 2022; Wolf et al., 2009). Overall, as the level of suspect resistance increases, officer use of force also increases, which aligns with departmental policies on use of force (Stoughton et al. 2020). Recent work also indicates that public perceptions of police use of force are impacted by a civilian's actions. McClure and colleagues (2023) examined how public perceptions of officer-civilian interactions may be affected by civilians' non-verbal actions (e.g., clenched fists, standing close to the officer) and found that civilians' non-verbal indicators of threat increased participants' perceptions that the encounter would turn violent. This provides initial evidence that civilians' situation-specific actions can impact public perceptions of police-civilian encounters.

Additional recent work exploring the impact of situation-specific circumstances on perceptions of use of force incidents suggests a significant impact of civilian action as well. Vardsveen and Wiener (2022) used a mock civil trial context to manipulate civilian threat as either ambiguous (suspect pulled out an unidentified object) or high (suspect pulled out a knife). Participants thought the officer's use of lethal force was more reasonable when suspect threat was high compared to low (Vardsveen and Wiener 2022). In conjunction with civilian threat, additional findings emphasize that individuals attend to civilian resistance when judging use of force encounters as well. Individuals perceive an officer's use of non-lethal force against a civilian as more reasonable when the civilian engages in assaultive behavior towards

the officer compared to active resistance (i.e., attempting to flee the scene; Flick and Schweitzer 2025). Taken together, these findings emphasize the importance of civilian behavior on the public's perceptions in use of force scenarios.

Current Study

Given recent emphases on perceptions of threat in use of force encounters (e.g., Bennell et al. 2021; Krajewski et al., 2024; McClure et al. 2023), the current study expands on previous research by examining the extent to which demographic and attitudinal factors predict perceptions of both threat to the officer and threat to the civilian in a non-lethal police use of force encounter. We also explore whether these demographic and attitudinal factors predict public perceptions of the threat at the beginning of an encounter (the initial threat) and the end of the encounter, while controlling for the situational-specific influence of civilian behavior. Lastly, although we were primarily interested in expanding on previous work by examining perceptions of threat specifically, we also measured perceptions of officer and civilian responsibility and reasonableness of force to examine alignment with previous work.

Research Questions (RQ):

RQ1: How does a perceivers' sex, race, age, political orientation, income, education, police legitimacy beliefs, and authoritarian views predict their perceptions of initial threat

RQ2: How does a perceivers' sex, race, age, political orientation, income, education, police legitimacy beliefs, and authoritarian views predict their perceptions of overall threat in a police use of force encounter?

RQ3: Does the impact of these demographic and attitudinal factors on threat perceptions vary when controlling for the impact of a civilian's actions during the encounter?

RQ4: How does a perceivers' sex, race, age, political orientation, income, education, police legitimacy beliefs, authoritarian views, and the situation-specific civilian behavior predict perceptions of officer responsibility, civilian responsibility, and reasonableness of force?

Hypotheses

We predict that participants will perceive (a) greater overall threat to the officer, (b) less overall threat to the civilian, (c) less officer responsibility, (d) more civilian responsibility, and (e) the officer's force to be more reasonable, when participants (H1) are White compared to non-White, (H2) are more conservative compared to more liberal, (H3) have

higher incomes compared lower incomes, (H4) believe the police are more legitimate compared to less, (H5) are more authoritarian compared to less, and (H6) when the civilian engages in more noncompliant actions compared to less.

Given the inconsistent previous findings related to participant sex, age, and education, we did not make specific a priori hypotheses for the impact of these predictors on any of our dependent measures. Further, given that to our knowledge previous work has not examined the impact that perceivers' demographics and attitudes have on initial perceptions of threat in a police-civilian encounter, we did not have a priori hypotheses for perceptions of initial threat to the officer or the civilian.

Method

Participants

Participants were recruited online through Prolific and included 724 community members over the age of 18 who passed all attention checks and reside in the United States. Participation took approximately 15 min, and participants were compensated \$2.05 USD. A post-hoc sensitivity analysis using the *pwr* package (Champely 2020) in R 4.3.1 (R Core Team 2023) indicated that we had 0.80 power for a minimum detectable effect size (f^2) of 0.02, suggesting that the study was sufficiently powered to detect small effects (Cohen 1988).

Materials

The university's institutional review board approved all study materials and measures. Materials and data are included in an Online Supplemental Materials (OSM) document hosted at the Open Science Framework website, accessible at <https://osf.io/5a3vr>.

Use of Force Scenario

An audio-recorded (~5 min) vignette was presented to participants with the accompanying transcript alongside. The vignette consisted of a description of a hypothetical police use of force scenario, which was based on a real-life traffic stop (*Rice v. Morehouse* 2021). Within the description, the officer pulls the civilian over for a minor traffic violation and suspicion of driving under the influence. During the encounter, the civilian ignores the officer's demands to provide their license and registration. Next, the civilian engages in a certain action against the officer (described below in the civilian action manipulation). The officer responds with non-lethal UOF against the civilian, in which they use a

takedown technique and hard fist strikes before ultimately handcuffing the civilian.

Civilian Action Manipulation

Across all conditions, the officer asked the civilian to exit the vehicle, but the civilian refused. The officer then attempted to open the driver's door, but the civilian attempts to keep the car door shut. The officer then extracts the civilian from the vehicle and attempts to arrest the civilian for obstruction and delay. Once out of the vehicle, the civilian engages in certain noncompliant actions based on the condition. In the less noncompliance condition, the civilian attempts to flee the scene by running away. In the more noncompliance condition, the civilian attempted to flee, but prior to doing so, they lunged towards the officer with clenched fists and struck them on the shoulder. A pilot study (described in the OSM) confirmed that individuals perceived the civilian's actions in the less noncompliant condition to be less forceful, threatening, and physically harmful to the officer.¹

Measures

Participant Demographics

Sex Participant sex was measured via the following item, "What is your sex?", with response options: male, female, other². The sample self-identified as 50.8% male and 49.2% female.

Age Participant age was measured via open-ended response to the question, "What is your age?". Participants ranged in age from 18 to 84 ($M=42.29$, $SD=14.60$).

Race Participant race was measured via the following question, "What of the following best describes your race?". The response options with the accompany sample percentages

were as follows: White (76.80%), Black (8.56%), Asian (6.08%), Hispanic (4.56%), and Other (4.00%).

Political Orientation Participants' political orientation was assessed via one item, "Which of the following best describes your political orientation?", measured on a 7-point scale (1 = *extremely liberal* to 7 = *extremely conservative*). The sample was slightly liberal leaning ($M=3.26$, $SD=1.76$).

Income Participants' income was measured via the following question, "What is your annual income?" with a 5-point increasing scale ($M=2.46$, $SD=1.19$) with following response options: less than \$25,000 per year, \$26,000 to \$45,000 per year, \$46,000 to \$80,000 per year, \$81,000 to \$150,000 per year, and more than \$150,000 per year.

Education Participants' education level was measured via the following question, "What is the highest year of education you have attained?" with a 5-point increasing scale ($M=3.64$, $SD=0.95$) with the following response options: less than high school, high school graduate, partial college or junior college degree, college degree, and graduate degree.

Participant Attitudes

Attitudes Towards Police Legitimacy Scale (APLS) Participants' trait-level attitudes towards police legitimacy were measured using a validated (Provenza 2021) short form of the Attitudes towards Police Legitimacy Scale (APLS; Reynolds et al. 2018). The scale contains 17 items that were summed to create a total ($\alpha = 0.99$) with higher scores indicating more positive attitudes towards the police ($M=69.52$, $SD=30.00$, range: 17–119). We chose the APLS instead of other assessments of police legitimacy because we were interested in the extent to which people perceive the police to be legitimate, not whether the police are, in fact, legitimate.

Authoritarianism Participants' trait-level authoritarian beliefs were assessed via the Aggression-Submission-Conventionalism (ASC) Scale (Dunwoody and Funke 2016). The ASC scale is a validated 18-item measure of authoritarianism derived from the three dimensions of the seminal Right Wing Authoritarianism scale (Altemeyer 1981, 1988) that includes the three original dimensions of authoritarian aggression, authoritarian submission, and conventionalism. ASC scale has shown greater divergent validity with political orientation by including items that are not as strongly linked to right-wing attitudes and behaviors (Dunwoody and Funke 2016; Torres-Vega et al. 2021). An example item

¹ Civilian injury severity (less v. more severe) was also manipulated within the vignette, with the full operationalization described in the OSM. Additional analyses indicated there were no significant interactions between civilian action and civilian injury on any of our dependent measures. In other words, the impact of civilian action was not dependent on civilian injury level. Since civilian injury was not of interest to our research question for the current manuscript, we do not discuss it any further.

² Seven participants self-identified as "other" and were removed from the sample to allow for analyses on the basis of sex.

includes “People in positions of authority generally tell the truth”. Participants’ scores were summed ($\alpha = 0.92$) with higher scores indicating stronger authoritarian beliefs ($M=60.75$, $SD=19.99$, range=18–120).

Dependent Measures

Initial Threat to the Officer and Civilian Participants’ perceptions of the extent to which the officer and the civilian were threatened at the beginning of the encounter prior to any civilian noncompliance or officer force was measured via one item each. The item was measured on a 9-point scale from 1 = *strongly disagree* to 9 = *strongly agree* and stated, “The officer [civilian] feared for his safety”. Perceptions of initial threat to the officer ($M=2.57$, $SD=1.97$) and threat to the civilian ($M=2.85$, $SD=2.02$) were measured directly after the first paragraph of the vignette, which described the civilian pulling over to the side of the road after seeing the officer’s police lights in his rearview mirror (see OSM for full description).

Overall Threat to the Officer and Civilian Participants’ perceptions of the extent to which the officer and civilian were threatened following exposure to the entirety of the use of force encounter were measured via one item each on a 9-point scale from 1 = *not at all* to 9 = *completely*. The item stated, “To what extent do you believe the officer [civilian] felt threatened in the situation?” ($M_{\text{officer}} = 5.30$, $SD_{\text{officer}} = 2.50$; $M_{\text{civilian}} = 5.44$, $SD_{\text{civilian}} = 2.50$).

Officer Responsibility Participants’ perceptions of the officers’ responsibility were measured via the following item, “To what extent is the officer responsible for the injuries sustained by the civilian?” on a 9-point scale (1 = *not at all* to 9 = *completely*; $M=5.38$, $SD=2.83$).

Civilian Responsibility Participants’ perceptions of the civilian’s responsibility were measured via the following item, “To what extent is the civilian to blame for the incident?” on a 9-point scale (1 = *not at all* to 9 = *completely*; $M=6.74$, $SD=2.32$).

Perceptions of Reasonable Force Two items assessed participants’ attitudes about whether the officer acted reasonably. The first item was, “To what extent do you believe the officer’s use of force against the civilian was reasonable?” (1 = *extremely unreasonable*, 9 = *extremely reasonable*). The second item was, “Do you believe that the level of force used by the officer was similar to the level of force that any reasonable officer on scene would have used?” (1 = *Not at all*, 9 = *Completely*). The attitudes were summed

to create a measure of participants’ perceptions of reasonable force ($r=.90$; $M=10.71$, $SD=5.18$).

Procedure

The study was conducted via Qualtrics. In the informed consent, participants were instructed they would hear about a brief police UOF scenario and would be asked to answer questions about the UOF incident, and complete general attitude measures and demographics. Participants were randomly assigned to one of two conditions that varied in civilian action (less noncompliance v. more noncompliance). After listening to the first paragraph of the scenario (which was consistent across conditions), participants completed the initial measures of threat. Next, participants listened to the remainder of the scenario description. Following this, participants completed their perceptions of reasonable force, which were followed by their perceptions of overall threat and responsibility in a randomized order. Participants then completed the two attitude measures (APLS and ASC) in a randomized order. Lastly, participants completed demographic information, were thanked for their time, and provided their code for payment.

Results

For each dependent variable, we report hierarchical regression models where the first step includes only participant demographic variables (i.e., sex, race³, age, political orientation, income, and education), the second step includes participant demographics and attitudes (i.e., APLS and ASC), and the third step includes participants demographics, attitudes, and our situational factor of civilian action (less noncompliance v. more noncompliance)⁴. Key findings are communicated in-text, and full model statistics are reported in Tables 1, 2, 3, 4 and 5. All analyses were performed using the statistical program R (R Core Team, 2024) and its base packages. The following additional packages were used to restructure and examine the data and create tables: *dplyr* (Wickham et al., 2023), *lsr* (Navarro 2015), *psych* (Revelle 2025), *sjPlot* (Lüdtke 2024), and *sur* (Harel 2020).

³ Given the large percentage of participants that identified as White (76.08%), race as a predictor was conceptualized as White participants compared to non-White participants (23.92%).

⁴ The situational variable of civilian action was not included in models predicting initial threat given that participant perceptions of initial threat were assessed prior to any manipulation of civilian action in the vignette.

Table 1 Multiple linear regression models with participant demographics predicting initial threat dependent variables

<i>Predictors</i>	Initial Threat to the Officer			Initial Threat to the Civilian		
	<i>Estimate</i>	<i>CI</i>	<i>p</i>	<i>Estimate</i>	<i>CI</i>	<i>p</i>
Intercept	1.57	0.59–2.55	0.002	3.35	2.35–4.35	<0.001
Sex	−0.04	−0.33–0.25	0.79	−0.04	−0.33–0.26	0.80
Race	−0.07	−0.24–0.10	0.41	−0.13	−0.30–0.05	0.16
Age	0.00	−0.01–0.01	0.94	−0.01	−0.02–0.00	0.12
Political Orientation	0.21	0.12–0.29	<0.001	−0.17	−0.25–0.08	<0.001
Income	−0.01	−0.15–0.12	0.83	−0.07	−0.21–0.06	0.29
Education	0.17	−0.00–0.34	0.05	0.27	0.10–0.45	0.002
<i>N</i>	724			724		
<i>R</i> ² / <i>R</i> ² adjusted	0.04/0.03			0.05/0.04		

Bolded *p*-values indicate *p*<.05

Table 2 Multiple linear regression models with participant demographics, APLS, and ASC predicting initial threat dependent variables

<i>Predictors</i>	Initial Threat to the Officer			Initial Threat to the Civilian		
	<i>Estimate</i>	<i>CI</i>	<i>p</i>	<i>Estimate</i>	<i>CI</i>	<i>p</i>
Intercept	1.07	0.06–2.08	0.038	3.66	2.64–4.69	<0.001
Sex	−0.06	−0.34–0.23	0.70	−0.02	−0.31–0.27	0.90
Race	−0.05	−0.22–0.12	0.55	−0.15	−0.32–0.02	0.08
Age	−0.01	−0.02–0.01	0.32	−0.00	−0.01–0.01	0.79
Political Orientation	0.07	−0.03–0.17	0.17	−0.05	−0.15–0.06	0.35
Income	−0.04	−0.17–0.10	0.58	−0.05	−0.19–0.09	0.47
Education	0.17	0.00–0.34	0.048	0.27	0.10–0.44	0.002
APLS	0.01	0.00–0.02	0.04	−0.01	−0.02–0.01	<0.001
ASC	0.01	−0.00–0.02	0.08	−0.00	−0.01–0.01	0.97
<i>N</i>	724			724		
<i>R</i> ² / <i>R</i> ² adjusted	0.07/0.06			0.08/0.07		

Bolded *p*-values indicate *p*<.05

Initial Threat To the Officer

The first step of the regression model ($r^2=0.04$, Table 1), including only participant demographics, indicated participant political orientation ($\beta=0.21$, $SE=0.04$, $p<.001$, $\eta^2=0.03$) and education ($\beta=0.17$, $SE=0.09$, $p=.05$, $\eta^2=0.01$) significantly predicted participants' perceptions of the amount of threat the officer was experiencing at the beginning of the encounter. More conservative and more educated participants reported the officer was more threatened compared to more liberal and less educated participants. The second step ($r^2=0.07$, Table 2), adding in participant attitudes, indicated the only significant predictors were participant education ($\beta=0.17$, $SE=0.09$, $p=.048$, $\eta^2=0.01$) and APLS ($\beta=0.01$, $SE=0.004$, $p=.04$, $\eta^2=0.01$). Again, participants with more, compared to less, education believed the officer experienced greater initial threat. Participants who believe the police to be more legitimate also believed the officer was more initially threatened, than those who believe the police to be less legitimate.

Initial Threat To the Civilian

The first step of the regression model ($r^2=0.05$, Table 1), including only participant demographics, indicated participant political orientation ($\beta=-0.17$, $SE=0.04$, $p<.001$, $\eta^2=0.02$) and education ($\beta=0.27$, $SE=0.09$, $p=.002$, $\eta^2=0.01$) significantly predicted participants' perceptions of the amount of threat the civilian was experiencing at the beginning of the encounter. More liberal and more educated participants reported the civilian was more threatened compared to more conservative and less educated participants. The second step ($r^2=0.08$, Table 2), adding in participant attitudes, again indicated the only significant predictors of were participant education ($\beta=0.27$, $SE=0.09$, $p=.002$, $\eta^2=0.01$) and APLS ($\beta=-0.01$, $SE=0.004$, $p<.001$, $\eta^2=0.02$). More (compared to less) educated participants and participants who believe the police to be less (compared to more) legitimate believed the civilian was more initially threatened.

Table 3 Multiple linear regression models with participant demographics predicting Post-Encounter dependent variables

Predictors	Threat to Officer			Threat to Civilian			Officer Responsibility			Civilian Responsibility			Reasonableness		
	Estimate	CI	p	Estimate	CI	p	Estimate	CI	p	Estimate	CI	p	Estimate	CI	p
(Intercept)	3.05	1.83–4.26	<0.001	6.85	5.64–8.06	<0.001	6.95	5.59–8.32	<0.001	5.56	4.48–6.65	<0.001	8.01	5.53–10.49	<0.001
Sex	0.07	–0.28–0.43	0.70	0.05	–0.30–0.41	0.76	0.21	–0.19–0.61	0.30	–0.23	–0.55–0.09	0.15	–0.57	–1.30–0.15	0.12
Race	–0.02	–0.23–0.19	0.82	0.01	–0.20–0.22	0.91	–0.06	–0.30–0.17	0.59	–0.07	–0.26–0.12	0.48	0.04	–0.39–0.47	0.84
Age	0.01	0.00–0.03	0.003	–0.00	–0.01–0.01	0.84	–0.02	–0.03–0.01	0.005	0.02	0.01–0.03	0.002	0.03	0.00–0.05	0.005
Political Orientation	0.38	0.27–0.48	<0.001	–0.41	–0.51–0.30	<0.001	–0.43	–0.54–0.31	<0.001	0.44	0.35–0.53	<0.001	0.85	0.64–1.06	<0.001
Income	0.15	–0.02–0.32	0.08	–0.15	–0.32–0.01	0.07	–0.16	–0.35–0.02	0.09	0.08	–0.07–0.23	0.30	0.32	–0.02–0.66	0.07
Education	0.02	–0.19–0.23	0.86	0.06	–0.15–0.27	0.55	0.25	0.02–0.49	0.04	–0.17	–0.36–0.01	0.07	–0.34	–0.77–0.08	0.12
N	724			724			724			724			724		
R ² /R ² adjusted	0.09/0.08			0.09/0.08			0.11/0.10			0.15/0.15			0.12/0.11		

Bolded p-values indicate $p < .05$ **Table 4** Multiple linear regression models with participant demographics, APLS, and ASC predicting Post-Encounter dependent variables

Predictors	Threat to Officer			Threat to Civilian			Officer Responsibility			Civilian Responsibility			Reasonableness		
	Estimate	CI	p	Estimate	CI	p	Estimate	CI	p	Estimate	CI	p	Estimate	CI	p
Intercept	1.77	0.62–2.91	0.003	8.22	7.04–9.40	<0.001	8.36	7.06–9.67	<0.001	4.15	3.15–5.15	<0.001	4.77	2.46–7.09	<0.001
Sex	0.01	–0.31–0.34	0.93	0.10	–0.24–0.43	0.57	0.27	–0.10–0.63	0.15	–0.29	–0.57–0.00	0.046	–0.68	–1.33–0.03	0.04
Race	0.05	–0.14–0.24	0.63	–0.04	–0.23–0.16	0.72	–0.14	–0.35–0.08	0.22	–0.00	–0.17–0.16	0.97	0.18	–0.21–0.56	0.37
Age	–0.01	–0.02–0.01	0.34	0.01	0.00–0.02	0.002	0.03	–0.01–0.01	0.97	–0.00	–0.01–0.01	0.90	–0.01	–0.03–0.01	0.34
Political Orientation	–0.02	–0.14–0.10	0.73	–0.04	–0.16–0.08	0.52	–0.00	–0.14–0.13	0.96	0.03	–0.07–0.13	0.59	–0.05	–0.29–0.18	0.66
Income	0.08	–0.07–0.23	0.31	–0.09	–0.25–0.06	0.24	–0.09	–0.26–0.08	0.32	0.01	–0.12–0.14	0.92	0.16	–0.14–0.47	0.29
Education	0.02	–0.17–0.21	0.83	0.06	–0.14–0.25	0.55	0.25	0.04–0.47	0.02	–0.17	–0.34–0.01	0.04	–0.33	–0.72–0.05	0.09
APLS	0.03	0.03–0.04	<0.001	–0.02	–0.03–0.01	<0.001	–0.03	–0.04–0.02	<0.001	0.03	0.02–0.04	<0.001	0.06	0.04–0.07	<0.001
ASC	0.02	0.00–0.03	0.003	0.01	–0.03–0.05	0.02	–0.02	–0.04–0.01	0.004	0.03	0.01–0.04	<0.001	0.07	0.04–0.09	<0.001
N	724			724			724			724			724		
R ² /R ² adjusted	0.26/0.25			0.21/0.20			0.24/0.24			0.34/0.34			0.29/0.29		

Bolded p-values indicate $p < .05$

Table 5 Multiple linear regression models with participant demographics, APLS, ASC, and civilian behavior predicting Post-Encounter dependent variables

Predictors	Threat to Officer			Threat to Civilian			Officer Responsibility			Civilian Responsibility			Reasonableness		
	Estimate	CI	p	Estimate	CI	p	Estimate	CI	p	Estimate	CI	p	Estimate	CI	p
Intercept	1.21	0.10–2.31	0.03	8.54	7.36–9.71	<0.001	8.77	7.48–10.07	<0.001	3.80	2.82–4.79	<0.001	4.05	1.75–6.34	0.001
Sex	0.08	–0.23–0.39	0.62	0.06	–0.27–0.39	0.72	0.22	–0.14–0.58	0.23	–0.25	–0.52–0.03	0.08	–0.60	–1.24–0.04	0.07
Race	–0.00	–0.19–0.18	0.97	–0.01	–0.20–0.19	0.94	–0.10	–0.31–0.12	0.37	–0.03	–0.20–0.13	0.68	0.11	–0.27–0.49	0.56
Age	–0.00	–0.01–0.01	0.68	0.01	–0.00–0.02	0.06	–0.00	–0.02–0.01	0.68	0.00	–0.01–0.01	0.78	–0.01	–0.03–0.02	0.54
Political Orientation	0.02	–0.09–0.13	0.69	–0.06	–0.18–0.06	0.29	–0.03	–0.17–0.10	0.60	0.05	–0.05–0.15	0.29	0.00	–0.23–0.24	0.98
Income	0.09	–0.06–0.23	0.24	–0.10	–0.25–0.05	0.20	–0.09	–0.26–0.07	0.27	0.01	–0.12–0.14	0.84	0.18	–0.12–0.48	0.25
Education	0.01	–0.18–0.19	0.95	0.07	–0.13–0.26	0.49	0.26	0.05–0.47	0.02	–0.18	–0.34–0.02	0.03	–0.35	–0.73–0.02	0.07
APLS	0.03	0.02–0.04	<0.001	–0.02	–0.03–0.01	<0.001	–0.03	–0.04–0.02	<0.001	0.03	0.02–0.03	<0.001	0.05	0.04–0.07	<0.001
ASC	0.02	0.00–0.03	0.003	–0.03	–0.04–0.02	<0.001	–0.02	–0.04–0.01	0.005	0.02	0.01–0.04	<0.001	0.06	0.04–0.09	<0.001
Civilian Behavior	1.24	0.93–1.55	<0.001	–0.70	–1.03–0.37	<0.001	–0.91	–1.27–0.55	<0.001	0.76	0.48–1.03	<0.001	1.61	0.97–2.25	<0.001
N	724			724			724			724			724		
R ² /R ² adjusted	0.32/0.31			0.23/0.22			0.27/0.26			0.37/0.36			0.32/0.31		

Bolder p-values indicate $p < .05$

Overall Threat To the Officer

The first step of the regression model ($r^2=0.09$, Table 3), including only participant demographics, indicated participant age ($\beta=0.01$, $SE=0.01$, $p=.03$, $\eta^2=0.01$) and political orientation ($\beta=0.38$, $SE=0.05$, $p<.001$, $\eta^2=0.07$) significantly predicted participants' perceptions of the amount of threat the officer was experiencing during the use of force encounter. Older and more conservative participants reported the officer was more threatened during the interaction with the civilian than younger and more liberal participants. The second step ($r^2=0.26$, Table 4), adding in participant attitudes, indicated the only significant predictors were APLS ($\beta=0.03$, $SE=0.004$, $p<.001$, $\eta^2=0.06$) and ASC ($\beta=0.02$, $SE=0.01$, $p=.01$, $\eta^2=0.01$). Participants who believe the police to be more legitimate and who are higher in authoritarianism perceived the officer to be more threatened in the use of force encounter, compared to those who believe the police to be less legitimate and are lower in authoritarianism. The third step of our model ($r^2=0.32$, Table 5) added in our manipulation of civilian action. Results indicated significant effects of APLS ($\beta=0.03$, $SE=0.004$, $p<.001$, $\eta^2=0.05$), ASC ($\beta=0.02$, $SE=0.01$, $p=.01$, $\eta^2=0.01$), and civilian action ($\beta=1.24$, $SE=0.16$, $p<.001$, $\eta^2=0.06$) on participants' perceptions of the amount of threat the officer experienced during the use of force interaction. Participants who heard about the civilian engaging in more, compared to less, noncompliance perceived that the officer experienced a greater threat during the encounter.

Overall Threat To Civilian

The first step of the regression model ($r^2=0.09$, Table 3), including only participant demographics, indicated only participant political orientation ($\beta = -0.41$, $SE=0.05$, $p<.001$, $\eta^2=0.08$) significantly predicted participants' perceptions of the amount of threat the civilian was experiencing during the use of force encounter. More liberal participants reported the civilian was more threatened, compared to more conservative participants. The second step ($r^2=0.21$, Table 4), adding in participant attitudes, indicated significant predictors of participant age ($\beta=0.01$, $SE=0.01$, $p=.03$, $\eta^2=0.005$), APLS ($\beta = -0.02$, $SE=0.004$, $p<.001$, $\eta^2=0.02$), and ASC ($\beta = -0.03$, $SE=0.01$, $p<.001$, $\eta^2=0.02$). Older participants, participants who believe the police to be less legitimate, and participants who are lower in authoritarianism indicated the civilian was more threatened during the use of force encounter, compared to younger participants, those who believe the police to be more legitimate, and those who are higher in authoritarianism. The third step ($r^2=0.23$, Table 5), adding in our independent variable of civilian action, again indicated significant effects of APLS ($\beta = -0.02$, $SE=0.004$, $p<.001$,

$\eta^2=0.02$), ASC ($\beta = -0.03$, $SE=0.01$, $p<.001$, $\eta^2=0.02$), and civilian action ($\beta = -0.70$, $SE=0.17$, $p<.001$, $\eta^2=0.02$). Participants who heard about the civilian engaging in more, compared to less, noncompliance perceived the civilian to be experiencing less threat during the use of force encounter.

Officer Responsibility

The first step of the regression model ($r^2=0.11$, Table 3), including only participant demographics, indicated participant age ($\beta = -0.02$, $SE=0.01$, $p=.005$, $\eta^2=0.01$), political orientation ($\beta = -0.43$, $SE=0.06$, $p<.001$, $\eta^2=0.07$), and education ($\beta=0.25$, $SE=0.12$, $p=.03$, $\eta^2=0.01$) significantly predicted perceptions of how responsible the officer was for the injuries sustained by the civilian. Younger, more liberal, and more educated participants reported that the officer was more responsible for the encounter, compared to older, more conservative, and less educated participants. The second step ($r^2=0.25$, Table 4), adding in participant attitudes, indicated significant effects of participant education level ($\beta=0.25$, $SE=0.11$, $p=.02$, $\eta^2=0.01$), APLS ($\beta = -0.03$, $SE=0.005$, $p<.001$, $\eta^2=0.05$), and ASC ($\beta = -0.02$, $SE=0.01$, $p=.004$, $\eta^2=0.01$) on perceptions of officer responsibility. Participants who believe the police to be less legitimate and who are lower in authoritarianism reported the officer was more responsible than those who believe the police to be more legitimate and who are higher in authoritarianism. The third step ($r^2=0.27$, Table 5), adding in our manipulation of civilian action, indicated significant effects of education ($\beta=0.26$, $SE=0.11$, $p=.02$, $\eta^2=0.01$), APLS ($\beta = -0.03$, $SE=0.005$, $p<.001$, $\eta^2=0.04$), ASC ($\beta = -0.02$, $SE=0.01$, $p=.005$, $\eta^2=0.01$), and civilian action ($\beta = -0.91$, $SE=0.18$, $p<.001$, $\eta^2=0.03$) on participants' perceptions of how responsible the officer was for the injuries sustained by the civilian. Participants who heard about the civilian engaging in less, compared to more, noncompliance perceived the officer as more responsible.

Civilian Responsibility

The first step of the regression model ($r^2=0.15$, Table 3), including only participant demographics, indicated participant age ($\beta=0.02$, $SE=0.01$, $p=.002$, $\eta^2=0.01$) and political orientation ($\beta=0.44$, $SE=0.05$, $p<.001$, $\eta^2=0.10$) significantly predicted participants' perceptions of how responsible the civilian was for the incident. Older and more conservative participants reported the civilian was more responsible, compared to younger and more liberal participants. The second step ($r^2=0.34$, Table 4), adding in participant attitudes, indicated significant impacts of participant sex, ($\beta = -0.29$, $SE=0.14$, $p=.046$, $\eta^2=0.004$), education level ($\beta = -0.17$, $SE=0.08$, $p=.04$, $\eta^2=0.004$), APLS ($\beta=0.03$, $SE=0.004$, $p<.001$, $\eta^2=0.05$), and ASC ($\beta=0.03$,

$SE=0.01$, $p<.001$, $\eta^2=0.02$). Male participants, less educated participants, participants who believe the police to be more legitimate, and those who are higher in authoritarianism indicated the civilian was more responsible, compared to more educated participants, those who believe the police to be less legitimate, and those who are lower in authoritarianism. The third step ($r^2=0.37$, Table 5), adding in our manipulation of civilian action, indicated significant effects of education ($\beta = -0.18$, $SE=0.08$, $p=.03$, $\eta^2=0.004$), APLS ($\beta=0.03$, $SE=0.004$, $p<.001$, $\eta^2=0.05$), ASC ($\beta=0.02$, $SE=0.01$, $p<.001$, $\eta^2=0.02$), and civilian action ($\beta=0.76$, $SE=0.14$, $p<.001$, $\eta^2=0.03$) on participants' perceptions of civilian responsibility. Participants who heard about the civilian engaging in more, compared to less, noncompliance perceived the civilian as more responsible for the incident.

Reasonableness

The first step of the regression model ($r^2=0.12$, Table 3), including only participant demographics, indicated participant age ($\beta=0.03$, $SE=0.01$, $p=.04$, $\eta^2=0.01$) and political orientation ($\beta=0.85$, $SE=0.11$, $p<.001$, $\eta^2=0.08$) significantly predicted participants' perceptions of the reasonableness of the officer's force. Older and more conservative participants thought the officer's force was more reasonable, compared to younger and more liberal participants. The second step ($r^2=0.29$, Table 4), adding in participant attitudes, indicated participants' perceptions of reasonableness were significantly predicted by participant sex ($\beta = -0.68$, $SE=0.33$, $p=.04$, $\eta^2=0.004$), APLS ($\beta=0.06$, $SE=0.01$, $p<.001$, $\eta^2=0.04$), and ASC ($\beta=0.07$, $SE=0.01$, $p<.001$, $\eta^2=0.02$). Male participant, participants who believe the police to be more legitimate, and participants who are higher in authoritarianism believed the officer's force was more reasonable, compared to female participants, those who believe the police to be less legitimate, and those who are lower in authoritarianism. The third step ($r^2=0.32$, Table 5), adding in our manipulation of civilian action, indicated significant effects of APLS ($\beta=0.05$, $SE=0.01$, $p<.001$, $\eta^2=0.04$), ASC ($\beta=0.06$, $SE=0.01$, $p<.001$, $\eta^2=0.02$), and civilian action ($\beta=1.61$, $SE=0.32$, $p<.001$, $\eta^2=0.02$) on participants' perceptions of how reasonable the officer's force was. Participants who heard about the civilian engaging in more, compared to less, noncompliance perceived the officer's force as more reasonable.

Discussion

As mentioned, understanding the impact of both situation-specific and individual difference-level factors on public perceptions of both officers and civilians in police use of

force encounters is important for predicting legal decision-making, individuals' likelihood to engage in lawful activity, and police training. When considering the influence of demographics, our findings contrast past work that has found a consistent impact of race (e.g., Boudreau et al. 2019) and income level (e.g., Carter and Corra 2016) on police use of force judgments. Thus, findings do not support H1a-e or H3a-e. One potential explanation for why race did not predict individuals' judgments in the current work is the racial composition of our sample. Approximately 75% of our sample identified as White, which limited the extent of meaningful comparisons we could examine across racial groups. We discuss this point further in our limitations section. Previous research has also indicated that individuals with higher incomes tend to hold more positive perceptions of police use of force. Yet, we found no significant impact of income on perceptions in our current work. It is possible that the impact of income affects more general judgments of police (e.g., Braga et al. 2014) or overall approval of force (e.g., Mourtgos and Adams 2020) but does not translate to more specific perceptions of threat, responsibility, or reasonableness, which were assessed in the current work.

Our results suggest that an individuals' age, education, and political orientation are the most consistent predictors of their perceptions of threat, responsibility, and reasonableness of force. Supporting H2a-e, participants who were more conservative, compared to liberal, perceived the officer as more threatened (initially and overall), less responsible for the incident, and perceived the officer's force as more reasonable. More conservative individuals also perceived the civilian as less threatened (initially and overall) and as more responsible for the incident. These findings are aligned with previous work suggesting conservative individuals have more positive perceptions of police and their use of force (e.g., Braga et al. 2014; Cao and Stack 2005; Trahan and Russell 2017) and extend this notion to perceptions of threat specifically in an officer-civilian encounter, as well to perceptions of civilians in these encounters.

Notably, however, the impact of political orientation on public judgments was eliminated after accounting for individuals' police legitimacy attitudes and authoritarian views. This suggests that individuals' attitudes (i.e., APLS and ASC) are more important in predicting their perceptions of threat, responsibility and reasonableness in police use of force incidents, when compared to their political orientation. It is possible that APLS and ASC attitudes encompass an individuals' broader political ideological belief than political orientation alone, as suggested by Gerber and Jackson (2017), and thus once accounted for, the impact of political orientation is diminished. Further, the greater influence of attitudinal measures compared to demographic characteristics on individuals' legally relevant judgments

aligns with past work on juror judgments (e.g., Lieberman & Sales, 2006). In other words, the juror decision-making literature suggests that jurors' attitudes, compared to their demographics, are more predictive of their case-related judgments and decisions.

In addition to political orientation, participants' age and education level significantly impacted multiple use of force-related judgments. Older individuals tended to perceive more overall threat to the officer, less overall threat to the civilian (even when controlling for APLS and ASC), perceived the officer as less responsible, the civilian was more responsible, and thought the officer's force was more reasonable. Further, an individuals' education level somewhat consistently predicted their perceptions. Individuals with more formal education perceived both the officer and civilian as more initially threatened (even when controlling for police legitimacy attitudes and authoritarian views). However, participant educational level did not impact perceptions of overall threat. It is possible that individuals with more formal education have greater knowledge of the threat dynamics and potential negative consequences that might arise from when a police and civilian make contact, and thus perceived *both* the officer and civilian to be more initially threatened in the encounter. However, after exposure to the full vignette, the influence of situation-specific information (e.g., the civilian actions) was of greater importance to their threat perceptions.

Interestingly though, when considering responsibility outcomes, individuals with more formal education perceived the officer as more responsible and the civilian as less responsible, even when controlling for participant attitudes and the situational influence of civilian action; yet, this did not translate to reasonableness judgments. This effect is notable, given it was the only demographic characteristic that remained significant across all outcomes when controlling for attitudes or a civilians' actions in the encounter. The strong impact of education on use of force related judgments aligns with previous work by Braga and colleagues (2014) which found that education was the only control variable that predicted attitudes, such that those with more formal education were more disapproving of an officer's force. Future research should further explore the relationship between perceiver education level and perceptions of responsibility in use of force encounters.

Previous research on the impact of age and education on public perceptions of police and police use of force has been mixed. Our findings suggest that both age and education have independent influences on these perceptions, which aligns with some work (e.g., Trahan and Russell 2017) but contrasts other work that has found no significant differences based on age (e.g., Mourtgos and Adams 2020; Simon et al. 2021) or education (e.g., Johnson and Kuhns 2009). As

education level increased, participants' perceptions of the officer become more negative and perceptions of the civilian more positive. These findings contrast with various studies that indicate more formal education is associated with more positive perceptions of police (e.g., Simon et al. 2021; Trahan and Russell 2017) but align with others that suggest the inverse (e.g., Braga et al. 2014; Mourtgos and Adams 2020). It is possible that the impact of education on perceptions depends on whether more general or situation-specific judgments are assessed. In other words, some past work has assessed use of force judgments in a general, hypothetical sense (e.g., "a policeman striking a citizen"; Mourtgos and Adams 2020; Trahan and Russell 2017), while the current study asked about a specific use of force that included additional contextual information. Future research should examine this possibility to better understand the impact of formal education on perceptions of police-civilian interactions.

It is also important to note that participant sex (male vs. female) was a significant predictor of perceptions of civilian responsibility and reasonable force judgments. Male participants thought the civilian was more responsible and the officer's force was more reasonable when controlling for participants' APLS and ASC scores. This trend aligns with previous work suggesting men have more positive perceptions of police use of force than women (e.g., Johnson and Kuhns 2009; Ratcliffe et al. 2015; Simon et al. 2021). Yet, the findings are noteworthy such that they suggest other perceptions beyond those related to perceptions of threat may vary by sex and ultimately impact reasonable force judgments. In other words, some scholars (e.g., Bennell et al. 2021) suggest threat perceptions are particularly important in judgments of whether force is reasonable. Yet, these findings suggest individuals' threat perceptions do not vary by sex, but sex still impacted their overall perceptions of whether force used was reasonable. Future research should explore this to determine additional related judgments that may vary by sex, and if these effects are dependent on the sex of the officer and civilian.

Beyond demographics, participants' attitudes towards police legitimacy and authoritarianism had a consistent influence of all judgments within our study (except for authoritarian views predicting initial threat), supporting H5a-e and H6a-e. The notion that those with more positive attitudes towards police legitimacy and more authoritarian views had more positive perceptions of the officer, more negative perceptions of the civilian, and were more supportive of the officer's force aligns with previous work (e.g., Celestin and Kruschke 2019; Gerber and Jackson 2017). Further, as mentioned, these attitudes appear to be more influential to judgments than participants' demographic characteristics.

Lastly, the situational influence of a civilian's actions consistently predicted participants' perceptions of threat, responsibility, and reasonableness of force. The notion that

a more (compared to less) noncompliant civilian results in more positive perceptions of the officer and their use of force aligns with previous work on public perceptions of officer force (e.g., Flick 2023), as well as the law governing use of force (e.g., *Graham v. Connor* 1989), and common police training (e.g., Nowicki, 2001; Terrill & Paoline, 2013; Wolf et al. 2009). In other words, a civilian's actions should be, and are, a significant predictor of whether an officer decides to use force, and the extent of force used (Boivin 2016; McLean et al. 2022). Thus, given the notion that a civilian's actions impacted public perceptions of the use of force encounter suggests alignment between police training, the law, and community perception.

Overall, participants' age, sex, and education level were the most consistent demographic predictors of use of force-related judgments in the current study. Yet, our findings suggest that the impact of these demographics is diminished once participants' police legitimacy and authoritarian attitudes are accounted for (except for the impact of education on responsibility judgments discussed above). In contrast to demographics, participant attitudes continue to have a significant impact on their perceptions of threat, responsibility, and reasonableness in use of force encounters, even when controlling for a civilian's specific noncompliant actions in that encounter. Therefore, it appears particularly important for scholars and criminal justice professionals to assess these attitudes.

Implications

The findings of the present research suggest that attorneys in police excessive use of force cases should measure or ask questions regarding potential jurors' attitudes towards police legitimacy or authoritarianism during voir dire. Alternatively, it may be wise for judges to provide jurors with additional instructions that discuss how individuals' pre-existing attitudes regarding police can bias their interpretations of police-civilian interactions, particularly perceptions of how much threat the officer faced in the encounter. If jurors are aware of the possibility that their personal biases can impact seemingly objective observations of police-civilian interactions, such as how much the officer was threatened, jurors may have a higher likelihood of being able to reduce the role of these biases in their legal decisions. Recent work on the impact of jury instructions aimed at drawing jurors' attention to personal biases (e.g., implicit racial bias) suggests jurors exposed to such instructions are more likely to discuss the issue of their own or others' bias in their deliberations (Lynch et al. 2022).

Aside from a legal decision-making standpoint, these findings may also be used to educate officers on civilian judgments, perhaps through trainings. If police are aware that

certain individuals perceive officers as more threatening in general, trainings may be developed to aid police officers in reducing civilians' initial threat perceptions from the outset of the encounter, perhaps by emphasizing police goals of understanding the situation at hand and ultimately providing help or service to civilians if needed. Training and implementation that emphasizes reducing perceptions of threat from the beginning of the encounter may aid in preventing a cycle of escalating threat from each side, potentially ultimately decreasing police use of force and improving police-community relations.

The present findings also suggest that professionals developing other types of police trainings (e.g., cultural sensitivity trainings, use of force policies) should account for the significant impact of public attitudes on perceptions and judgments of the police. Not only do individual demographic characteristics influence how people view the police, but our findings and other research (e.g., Celestin and Kruschke 2019; Ewanation et al. 2022; Flick and Schweitzer 2025) continue to show that individuals' perceptions of police legitimacy shape how interactions with the police are perceived. Thus, we encourage agencies to take a multi-faceted approach when training officers, focusing not only on ensuring officer and community safety but also on understanding how individuals' demographics and views towards the police shape how civilians interact with the police. Improved police-community relations could have numerous downstream consequences, including (but not limited to) increased perceptions of police legitimacy, more positive interactions with police, less police use of force, and less biased applications of the reasonable officer standard at trial.

Limitations and Future Directions

The current work has several notable limitations. Although based on a real-life traffic stop, we used an audio-recorded vignette with an accompanying written transcript in our study, which may not elicit the same perceptions as viewing a police-civilian encounter occurring in real-time or viewing a video recording of the encounter. Future researchers should examine whether these factors predict public perceptions of threat when watching a police use of force incident. Another limitation is our use of a self-report methodology, particularly when assessing attitudes and perceptions of a publicly charged topic. Our participant responses may have been influenced by social desirability bias or participants' subjective interpretations especially given the potentially sensitive nature of the judgements assessed (Robinson and Clore 2002). Future work should consider the use of social desirability scales, such as the Marlowe-Crowne Social Desirability Scale (Crowne and Marlowe 1960), to explore the potential impact social desirability concerns had on our findings. Lastly, our work is limited by our primarily White participant sample.

This notably hinders the generalizability of our current findings to other racial groups. Further, as mentioned, our results did not suggest a significant impact of perceiver race on their use of force-related perceptions, which contrasts previous research. It is possible that these null findings were driven by our majority White sample and our use of a binary racial comparison (White v. non-White). Future work should aim to actively recruit non-White participants to determine whether our results hold with a more racially diverse sample.

Author Contributions C.F. conceptualized the study and design. Material preparation and data collection were performed by C.F. and K.S. Data analysis was performed by K.S. All authors prepared the first draft of the manuscript, and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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Data Availability Materials and data are included in an Online Supplemental Materials (OSM) document hosted at the Open Science Framework website, accessible at <https://osf.io/5a3vr>.

Declarations

Ethical Approval The university's institutional review board approved all study materials and measures (Protocol #20220615CF03332).

Competing interests The authors declare no competing interests.

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